

1. Overview

This section presents the detailed results of the Quality Assessment (QA) phase conducted for the **117 studies** identified for full-text review. Each study was evaluated against five rigorous technical criteria (QA1–QA5) specifically designed to measure the study's ability to answer the research questions regarding dataset provenance, feature engineering, and automated representation learning.

2. Scoring Methodology

To ensure methodological consistency and high-validated evidence (IC-5), scoring was performed using a standardized scale:

- **1.0:** Criterion is fully and explicitly addressed with technical detail.
- **0.5:** Criterion is partially addressed or lacks specific algorithmic formulas.
- **0.0:** Criterion is not addressed or is irrelevant to the technical scope.

Inclusion Threshold: A minimum cumulative score of **3.5 out of 5.0** was required for final inclusion.

#	Author (Year)	QA1: Dataset (Justification)	QA2: Preprocessing (Justification)	QA3: Expert Indicators (Justification)	QA4: Automated Architecture (Justification)	QA5: Results (Justification)	Total	Decision
1	Adun (2023)	1.0: Explicitly identifies the CERT Insider Threat dataset from Kaggle (v1). (Sec 3.5)	1.0: Details feature expunging of redundant columns and SVM labeling for unlabeled data. (Sec 3.5)	0.5: Lists 6 attributes (User, Action, etc.) but lacks deep logic on why these represent behavior. (Table 1)	1.0: Detailed 6-layer ANFIS structure with Bell membership functions explained. (Sec 3.2.2)	1.0: Reports 92% SVM accuracy and 91% ANFIS accuracy with error rates. (Sec 4)	4.5	Include
	Agarkar (2025)	1.0: Identifies use of anonymized logs , incident reports, and interviews from critical infrastructure sectors.	0.0: Mentions "Preprocessing and Data Cleaning" as a stage but does not provide technical algorithms for cleaning, noise reduction, or normalization.	0.0: Mentions " Expert Interviews " as a source but does not explicitly list the resulting indicators or rules used for the baseline.	1.0: Explains a framework using ML clustering (K-means, DBSCAN) and LSTM neural networks .	1.0: Reports a 92% TPR , 10% FPR, and 30% reduction in detection time.	3.0	Exclude
	Ahmad (2021)	0.5: Uses the NSL-KDD network intrusion dataset. While identified, it reflects low-level network traffic rather than the organizational user behavior required by IC-1.	0.0: Describes standard normalization but lacks specific "User Behavior" cleaning (e.g., session reconstruction or identity mapping) essential for this SLR.	0.5: The paper relies on raw network flow features (TCP/UDP); it does not define any expert human-behavioral rules or manual indicators. (Sec 5.1)	1.0: Explains supervised ensemble methods and SGD-based learners for anomaly detection. (Sec 4)	1.0: Reports standard metrics (Accuracy, F1, AUC) for 15 algorithm variations. (Table 5.2)	3.0	Exclude
2	Ahmad W (2025) Top 10 Cases	1.0: Identifies data collected from a live network	1.0: Details Data Cleaning (Wazuh manager rules),	1.0: Explicitly lists 10 prioritized use cases mapped to MITRE	1.0: Detailed explanation of the Random Cut Forest (RCF) unsupervised algorithm,	1.0: Reports TPR, FPR, Precision, F1-Score, and	5.0	Include

		environment at the Canadian Institute of Technology. Specifies 60-90 days of logs from Windows 11/2022 and Wazuh. (Sec. "Methodology", Page 280)	Feature Selection (e.g., LogonType, IpAddress), and Data Transformation (categorical to numerical) via OpenSearch plugins. (Sec. "Methodology", Page 280)	ATT&CK tactics (e.g., T1010, T1008) and provides specific Windows Event IDs (4624, 4672, etc.) as indicators. (Table 1 & Table 3)	including its integration into the Wazuh/OpenSearch SIEM for real-time streaming. (Sec. "Methodology", Page 280-281)	Detection Time (under 3 min for 80% of cases) across all 10 use cases. (Table 4, Page 289)		
3	Ahmadi (2025) Zero Trust	1.0: Identifies the use of anonymized enterprise logs and historical incident data representing distributed office environments. (Abstract & Sec 3.1)	0.5: Mentions data cleaning and normalization for contextual features, but lacks the specific algorithmic formulas for the cleaning phase. (Sec 3.1)	1.0: Explicitly defines a weighted risk-scoring model [Eq. 1] using expert-defined factors: Login Time (\$w_1\$), Geolocation (\$w_2\$), and Resource Sensitivity (\$w_3\$) . (Sec 1.2)	1.0: Detailed explanation of an AI-driven framework using Clustering (K-Means) for behavioral baselining and Random Forest for dynamic permission adjustment. (Sec 3.2)	1.0: Reports precision, scalability, and specific risk-reduction impacts (e.g., real-time isolation of compromised identities). (Abstract & Sec 4)	4.5	Include
4	Alabdulkreem (2022) weighted fusion	1.0: Identifies use of a dataset comprising 5000 samples (malicious vs. non-malicious). It aggregates five log types: removable device, logon/off, email, file, and HTTP. (Sec. "The proposed model" & Table 1)	1.0: Detailed steps provided for data aggregation (merging 5 files), feature extraction , and encoding for ML compatibility. (Sec. "Pre-processing and feature extraction")	1.0: Explicitly describes manual indicators: number of removable device uses outside office hours , visits to " wikileaks.org ", and logins outside regular hours . (Sec. "Pre-processing and feature extraction")	1.0: Explains a complex Weighted Fusion architecture combining LSTM, GRU, and Stacked Autoencoder (SAE) , with Ant Lion Optimizer (ALO) for hyperparameter tuning. (Sec. "Ensemble weighted fusion based classifier")	1.0: Reports detailed metrics including 99.40% Accuracy , Precision, Recall, F-Score, AUC, and ROC analysis with 70:30 and 80:20 splits. (Sec. "Experimental validation" & Table 4)	5.0	Include
5	Ali (2025)	1.0: Identifies a unique dataset of 1,306 expert-adjudicated cases from the US Army Insider Threat Hub (InT Hub). (Sec 2.1)	1.0: Details a Python-based script for extracting narrative text, dropping missing data, and performing Random Under-sampling (negligible class) and Over-sampling (medium class) to balance the corpus. (Sec 2.1 & 2.3)	1.0: Specifically describes the use of Dispositional and Situational traits as baseline indicators. It uses expert human analyst assessments (negligible to high) as the ground truth. (Sec 1.2 & 2.1)	1.0: Explains a complex NLP architecture using BERT (bidirectional encoder representations) , BERTopic , and Zero-shot topic modeling to extract latent semantic patterns from narrative reports. (Sec 2.2)	1.0: Reports a 96% Detection Accuracy Rate (DAR) , a metric designed to account for both accurate predictions and prudent over-estimations of threat. (Sec 2.5 & 3.3)	5.0	Include
6	Al-Mhiqani (2021) Imbalanced Learning	1.0: Explicitly identifies the CERT r4.2 dataset , detailing the 1,000 employees, 17-month duration, and 3 specific threat scenarios. (Sec III-A)	1.0: Detailed steps provided for Log Aggregation, Parsing (text to numeric), and scaling via MaxAbsScaler (range [-1,1]). Crucially, it details the use of ADASYN for	0.5: Identifies the 5 activity types (logon, device, email, HTTP, file) as the baseline but lacks a detailed list of the specific manual rules/expert logic used to categorize these as	1.0: Detailed explanation of the AD-DNN architecture, a Deep Neural Network with 20 hidden layers , utilizing the Adam optimizer and a specific batch size (\$1024 \times 16\$). (Sec III-C & IV)	1.0: Reports Accuracy (96%), F-Score (95%), AUC (95%), FPR (4%), and FNR (5%), providing a comparative table against SVM, LSTM,	4.5	Include

			synthetic oversampling. (Sec III-B, C)	"anomalous" prior to training. (Sec III-A)		and other baselines. (Table I)		
7	Al-Mhiqani (2022) new intelligent multilayer framework	1.0: Explicitly identifies the CERT r4.2 dataset and details the three specific insider attack scenarios used for evaluation. (Sec 5.2 & 5.3)	1.0: Describes a comprehensive preprocessing phase including Log Aggregation, Parsing (text to numeric), and feature space organization for 9 ML models. (Sec 4 & 5.1)	1.0: Detailed list of features and their value spaces is provided in Table 5, derived from expert-defined activities (logon, device, email, HTTP, file). (Sec 5.2, Table 5)	1.0: Explains a unique Multilayer Framework : Layer 1 uses Entropy-VIKOR for model selection; Layer 2 implements a Hybrid (HITD) model combining Random Forest (Misuse) and KNN (Anomaly) . (Sec 4 & 6.2)	1.0: Reports Accuracy (99% for known, 97% for unknown), FPR (0.29%), FNR, Precision, Recall, F-score, and AUC across multiple test sets. (Sec 6.2.1, Table 10)	5.0	Include
8	Almusawy (2024)	1.0: Explicitly identifies CERT r4.2 as the primary dataset, specifically utilizing the 1,000 employees and the subset of 70 insiders for experimentation. (Sec III-A)	1.0: Details a dual preprocessing approach: NoSQL-based aggregation for feature extraction and SMOTE (Synthetic Minority Oversampling Technique) to balance the 1000-record reduced set. (Sec III-B)	1.0: Provides a clear list of expert-defined features: AfterEnd (logon outside hours), BlockedSite (visited restricted URLs), CountActivity (external device usage), and No_of_Email (blocked email counts). (Table IV)	1.0: Explains an integrated framework that intersects Expert Policy Algorithms (if-then logic) with ML classifiers (XGBoost, AdaBoost, SVM, NB). (Sec III-B-3 & Algorithm 1)	1.0: Reports Accuracy, Precision, Recall, and F1-Score. Specifically highlights achieving 99% accuracy with XGBoost and 100% after intersection with expert rules. (Sec IV, Table VI)	5.0	Include
	Alnaim (2025)	0.5: Uses a hybrid network dataset (6G Synthetic + UNSW-NB15) centered on network traffic rather than organizational UBA logs.	0.5: Describes general network normalization but lacks specific behavioral feature engineering (email, file access) required for your ITD scope.	0.0: Trust scores are based on hardware/network entities; lacks exclusive focus on human behavioral rules/indicators.	1.0: Strong explanation of a 12-layer AI and Blockchain-based framework for decentralized security.	1.0: Reports clear performance metrics for 6G latency, throughput, and general attack mitigation.	3.0	EXCLUDE
9	Al-Shehari and Alsawail (2021) One-Hot Encoding	1.0: Utilizes the CERT R4.2 dataset , specifically focusing on the "dense needle" version involving 1,000 users over 17 months. (Sec 3.1)	1.0: Extensive preprocessing including Aggregation, One-Hot Encoding (for categorical variables), Feature Scaling (Standard Scalar), and SMOTE for class imbalance. (Sec 3.2-3.4)	1.0: Defines specific indicators based on threat scenarios: logins outside regular hours, USB usage outside office hours , and visits to wikileaks.org . (Sec 3.3)	1.0: Implements and compares a suite of ML algorithms (LR, DT, RF, NB, k-NN, KSVM) with detailed logic on how One-Hot encoding prevents misinterpretation of categorical correlations. (Sec 3.5)	1.0: Reports AUC-ROC of 0.99 , Precision, Recall, and F-measure. Provides comparative analysis against existing state-of-the-art models. (Sec 4.3 & Table 12)	5.0	Include
10	Alshehri Abdullah (2022) Multi-Sequence Representation	1.0: Utilizes the CERT dataset , specifically processing 32 million events (log-in, log-off, device, and HTTP) for more than 1,000 users. (Sec IV-A)	1.0: Describes a unique Multivariate Time Series representation. Details the conversion of user actions into ordered sets of real variables and the use of Sequence Activity Embedding to vectorize behavior. (Sec III-A, B)	0.5: Identifies the 4 primary log types as the source but relies on the latent temporal relationships extracted by the model rather than explicitly defining manual threshold-based rules. (Sec III-C)	1.0: Deeply explains a Relational Recurrent Neural Network (Rel-RNN) using LSTM units. Features a novel Direct Graph Architecture to learn inter-dependencies between different activity sequences. (Sec III-C, D)	1.0: Reports AUC of 0.99 and provides detailed performance metrics (F1-score, Precision, Recall) across various hidden-layer sizes (\$h=16\$ to \$h=256\$). (Sec IV-B)	4.5	Include

11	Al-Shehari T et al (2023) Isolation Forest	1.0: Utilizes the CERT r4.2 benchmark dataset , explicitly mentioning the 1,000 users and 500-day log span. (Sec III-A)	1.0: Detailed preprocessing steps: log consolidation, feature extraction, label encoding, and scaling . Uniquely addresses imbalance at the algorithm level rather than just data level. (Sec III-A, B)	1.0: Identifies manual behavioral indicators: logons outside office hours, USB usage counts, and WikiLeaks URL visits . (Sec III-A)	1.0: Implements the Isolation Forest (IF) algorithm, an unsupervised anomaly detection method that isolates outliers via random partitioning (path length) rather than profiling normal behavior. (Sec III-B-2)	1.0: Reports an F-score of 0.99 and Accuracy of 98% . Provides a comparative analysis against supervised baselines (XGBoost, RF, DT, KNN). (Sec IV-B, Table 7)	5.0	Include
12	Al-Shehari and Alsawail (2023) Random resampling	1.0: Utilizes the CERT r4.2 benchmarking dataset, specifically the data exfiltration scenario involving 1,000 employees over 17 months. (Sect 3.1)	1.0: Detailed feature engineering: log consolidation, Label Encoding, Standard Scaling , and a comprehensive comparison of Under-sampling, Over-sampling, and Hybrid-sampling techniques. (Sect 3.2-3.3)	1.0: Defines expert indicators for "Quitter" behavior: logins/removable device usage outside office hours and visits to wikileaks.org . (Sect 3.1)	1.0: Implements and compares XGBoost, Random Forest, Decision Tree, and KNN models. The "Automated" focus is on how random resampling transforms the data space for these algorithms. (Sect 3.4)	1.0: Reports ROC-AUC of 0.94 for DT with Under-sampling and 0.87 for KNN with Over-sampling. Provides comparison with baseline (imbalanced) results. (Sect 5, Table 10)	5.0	Include
13	Al-Shehari et al (2024) comparative evaluation + CNN	1.0: Analyzes the CERT dataset , specifically R4.2, which reports users' IT activities (login, HTTP, email, etc.) for over 1,000 employees. (Abstract & Intro)	1.0: Provides a rigorous comparison of three synthetic balancing techniques: SMOTE, Borderline-SMOTE, and ADASYN . Includes data transformation for CNN compatibility. (Abstract & Sect. 3)	1.0: Focuses on human behavior-based data (IT activities). While using automated CNNs, the study grounds the analysis in organizational behavior categories (malicious vs. negligent).	1.0: Develops a CNN-based detection model. Explains the integration of deep learning layers with sampling algorithms to handle high-dimensional, imbalanced behavioral data. (Abstract & Keywords)	1.0: Reports a ROC curve of 96% for the CNN + ADASYN hybrid model, explicitly comparing it to CNN + SMOTE and CNN + Borderline-SMOTE. (Abstract & Results)	5.0	Include
14	Al-Shehari et al (2024) LOF	1.0: Utilizes the CERT r4.2 dataset , simulating 1,000 employees over 500 days with over 32 million activities. (Sect. III-A)	1.0: Exhaustive preprocessing: Handling missing values (mean/mode imputation), One-Hot Encoding, Min-Max Scaling , and data partitioning. (Sect. III-A-1 to 5)	1.0: Explicitly targets the "Quitter" behavior: high-frequency USB usage after hours and WikiLeaks visits by users who previously showed different patterns. (Sect. III-A)	1.0: Implements the Density-Based Local Outlier Factor (DBLOF) . Provides mathematical proofs for Local Reachability Density (LRD) and the LOF score calculation. (Sect. III-B)	0.5: Reports an F-1 score only of 98% (at a 0.02 contamination rate). Provides detailed performance metrics across various contamination ratios. (Sect. IV-B, Table 1)	4.5	Include
15	AlSlaiman (2023)	1.0: Utilizes CERT r4.2 , representing 18 months of activity logs for 1,000 employees. (Abstract)	1.0: Implements a highly specialized pipeline: Sentiment Analysis for classification and Gray Encoding to maintain temporal relationships between activities. (Abstract & Sect. 4-7)	0.5: Uses sentiment analysis to classify activities into behavioral categories (benign vs. malicious) but focuses more on the encoding of these events than manual threshold rules. (Abstract)	1.0: Proposes a novel Long Short-Term Memory (LSTM) deep learning network designed to process variable-length samples. (Abstract & Sect. 8)	1.0: Reports an AUC of 97% , False Positive Rate of 0.29% , and False Negative Rate of 2.47% . (Abstract)	4.5	Include

16	Amiri-Zarandi (2023) FL	0.5: Validates the model using the CERT r4.2 dataset , specifically analyzing 1,000 users over 17 months. (Abstract & Sect. 4) Does not cover FE in details	1.0: Implements a complex pipeline for Federated Learning (FL) , including local model training and global aggregation to ensure Privacy . (Sect. 3.1)	0.5: Focuses on Explainable AI (XAI) using SHAP values to identify which features contribute to threat scores, rather than relying on manual expert-defined thresholds. (Sect. 3.2)	1.0: Proposes a decentralized architecture using Deep Autoencoders and GRUs (Gated Recurrent Units) within a Federated framework. (Sect. 3)	1.0: Demonstrates that the Federated approach outperforms individual local models in terms of F-measure and AUC while maintaining data privacy. (Abstract & Sect. 4.2)	4.0	Include
17	Amuda (2022)	1.0: Utilizes the CERT Insider Threat Synthetic dataset v4.2 , processing 3,015,990 log lines for 1,000 users. (Sect. 3.2)	1.0: Employs Min-Max normalization , Categorical Attributes Encoding , and Mutual Information Gain for dimensionality reduction/feature selection. (Sect. 3.2)	0.5: Identifies specific activities like after-hours logon and unauthorized file access (e.g., .exe, .zip) to build user profiles, though it relies on automated classification for final detection. (Sect. 3.2)	1.0: Formulates a hybrid model using the synergism of Convolutional Neural Network (CNN) and Gated Recurrent Unit (GRU) . (Sect. 3.3)	1.0: Reports a detection accuracy of 97.39% , sensitivity of 97.37% , and an AUC of 0.90 . (Sect. 4.2-4.3)	4.5	Include
18	Anakath (2022) Cloud	1.0: Uses open-source datasets and specific user interaction logs (mouse/keystrokes) evaluated via Cooja simulator . (Sect. 4, Page 486)	0.5: Mentions feature extraction for mouse/keystroke dynamics but lacks detailed data cleaning/normalization steps. (Sect. 3.1)	1.0: Identifies specific behavioral biometrics: mouse movement angles, click frequency, keystroke duration, and latency . (Sect. 3.1, Table 1 & 2)	1.0: Proposes a Deep Belief Neural Network (DBN) using stacked Restricted Boltzmann Machines (RBM) . (Sect. 3.2, Page 484)	1.0: Reports an accuracy of 99% and F-Measure of 98% , with comparisons to SVM and LSTM. (Sect. 4, Page 489)	4.5	Include
19	Anju (2024)	1.0: Utilizes both CERT R4.2 (1,000 users) and R6.2 (4,000 users) benchmark datasets. (Sect. 5.1)	1.0: Detailed steps including Z-score normalization for outliers, mean value imputation for missing data, and One-Hot/Integer encoding for features. (Sect. 4.1, 4.3)	1.0: Identifies specific behavioral markers: working after hours, WikiLeaks visits, job searching on rival sites , and abnormal USB/external hard drive usage . (Sect. 4.2)	1.0: Proposes a Stacked CNN (VGG19 + Xception) combined with an Attentional Bi-directional Gated Recurrent Unit (BiGRU) . (Sect. 4.5)	1.0: Reports 92.52% accuracy, 98% Precision, 95% Recall, and 96% F1-score with a comprehensive comparison against 10+ baselines. (Table 3, Sect. 5.3)	5.0	Include
20	Asha (2024)	1.0: Utilizes the CERT 3.2 synthetic dataset from Carnegie Mellon University, a standard benchmark in the field. (Abstract & Sect. 3.1)	1.0: Defines a comprehensive Data Sampling System comparing 8 techniques (ROS, SMOTE, ADASYN, RUS, ENN, NM-1, NM-2, T-L). (Sect. 4.1.3)	1.0: Identifies manual features such as Logon, Device, and HTTP records, specifically categorized as Vector, User, PC, and Activity. (Sect. 4.1.1 & 4.1.2)	1.0: Proposes a Novel Double-Layer Architecture : Layer 1 for data sampling/balancing and Layer 2 for anomaly detection. (Abstract & Sect. 1)	1.0: Reports clear metrics including 100% Recall and 78.72% F-score using One-class SVM with NM-2 sampling. (Abstract)	5.0	Include
21	Budžys (2024)	1.0: Validated on two publicly available benchmark collections: CMU Keystroke Dynamics	1.0: Implements Z-score for outlier removal and the novel Gabor Filter Matrix Transformation (GFMT) to convert 1D	1.0: Identifies specific manual features: Hold Time (HT) , Key Press Time (KPT) , and Key Interval Time (KIT) (Sect. 3.1).	1.0: Employs a Siamese Neural Network (SNN) with dual CNN branches for behavioral comparison (Sect. 3.2).	1.0: Reports an Equal Error Rate (EER) of 0.04545 , outperforming state-of-the-art image-transformation methods (Abstract).	5.0	Include

		and GREYC-NISLAB (Sect. 4.1).	data to images (Sect. 3.1).					
22	Cai (2024)	1.0: Validated on two standard benchmark datasets: CERT r4.2 and r5.2 . (Abstract, Sect. IV-A.1)	1.0: Defines activity coding ($\$ci = type(ai) \times 24 + time(ai)\$$), aggregation of multi-source logs, and session splitting (logon-to-logoff). (Sect. III-B.1, IV-A.2)	1.0: Identifies specific attack vectors (exfiltration, deletion, password compromise) and activity types (logon, file access, device connection). (Sect. II-B, III-B.1)	1.0: Proposes the LAN framework combining sequence encoders, graph structure learning for adaptive neighbors, and GNNs. (Sect. III, Fig. 2)	1.0: Reports AUC, DR, and FPR across 9 real-time and 8 post-hoc baselines, showing an improvement of at least 8.43% in AUC. (Sect. IV-B, Table II & III)	5.0	Include
23	Ding (2024)	1.0: Utilizes the CERT v4.2 benchmark dataset (1,000 users, 17 months of logs). (Sect. 5.1)	1.0: Implements seven heuristic rules for graph construction, activity type mapping (0–4), and a graph node elimination technique for normalization. (Sect. 3.1, 4.1, 4.2)	1.0: Defines explicit rules for Logon patterns (working vs. off-hours), Keyword blacklists (e.g., wikileaks.org), and Session statistics . (Sect. 4.1.3, 4.1.6)	1.0: Proposes ASG-ITD , utilizing Graph Convolutional Networks (GCN) on multi-level associated session graphs. (Sect. 4.4)	1.0: Reports 99% Accuracy and 99% F1-score , with a comprehensive ablation study and baseline comparison. (Abstract, Sect. 5.2)	5.0	Include
24	Dong (2025)	1.0: Evaluated on CMU CERT r5.2 (dense) and r6.2 (sparse) datasets. (Sect. V-A)	1.0: Features a User Context Feature Fusion module using Temporal Convolutional Networks (TCN) with dilated causal convolutions. (Sect. IV-C)	1.0: Maps behavioral deviations such as unauthorized access , privilege escalation , and atypical data transfer patterns across session, day, and week granularities. (Sect. I, V-A)	1.0: Proposes the DDCC pipeline , which synergizes Denoising Diffusion Probabilistic Models (DDPM) with Curriculum Learning (CL) . (Sect. IV-A)	1.0: Achieves AUC scores exceeding 0.97 across most granularities, outperforming 8 baselines including LAnoBERT and TCF-Trans. (Table IV, Sect. V-D)	5.0	Include
25	Dosh (2021)	1.0: Utilizes the CERT benchmark dataset (versions R4.1 and R4.2) featuring 1,000 employees over 17 months. (Sect. 4.1)	1.0: Clearly defines categorical-to-numeric encoding and Min-Max Normalization including the linear transformation formula. (Sect. 4.3 & 4.4)	0.5: Manually identifies five key features (Date, User, Source, Action, Vector) for the model, though it lacks the complex "behavioral rules" (e.g., specific time thresholds) found in other studies. (Sect. 4.3)	1.0: Implements three classic ML models: Random Forest , Naive Bayes , and 1-Nearest Neighbor (1-NN) , with mathematical descriptions of distance metrics. (Sect. 4.5)	1.0: Reports Accuracy, Error Rates, RMSE , and MAE for all models using both 66% splits and 10-fold cross-validation. (Sect. 4.5.4 & 5)	4.5	Include
26	Duan (2024)	1.0: Utilizes the CMU-CERT Insider Threat Dataset (r4.2) for training and validation. (Section 4.1)	1.0: Employs NLP techniques to scale content maliciousness and a Sparse Autoencoder (SAE) to transform numerical vectors into 32×32 images. (Section 4.1, 4.2)	1.0: Incorporates psychological traits (Big Five personality scores) and LDAP role features via one-hot encoding. (Table 2, Section 4.5)	1.0: Designs an Attention module for contextual extraction and uses a pre-trained ResNet with multi-source feature fusion. (Abstract, Section 4.2)	1.0: Reports an accuracy of 99.21% and an AUC of 0.9951 , outperforming multiple state-of-the-art baselines. (Table 5)	5.0	Include
27	Erola (2022)	1.0: Uniquely utilizes real-world data from three multinational	0.5: Describes data normalization and transformation into a	1.0: Explicitly focuses on human behavioral aspects and defines	1.0: Details the CITD tool architecture , which employs a multi-algorithm anomaly detection	1.0: Reports performance based on real-world incident	4.5	Include

		organizations (Financial, Telecommunications, and IT sectors) over a 12-month period.	common format but lacks the granular technical step-by-step formulas (e.g., specific scaling parameters) found in laboratory-only papers.	specific "indicators of interest" validated by organizational security analysts.	approach (Statistical and Machine Learning) tailored for real network infrastructure.	validation, discussing True/False positives and negatives in an operational context over 6 months of live deployment.		
28	Fei (2023)	1.0: Uses both CERT R4.2 and LANL .	1.0: Provides explicit formulas for word embeddings (Glove), cosine similarity, and text-vector summation	0.5: Defines three patterns: Attention, Emotion, and Behavior anomalies.	1.0: Proposes a clear unsupervised framework using Isolation Forest and ancillary risk factors.	1.0: Reports comprehensive metrics: AUC (0.93), Recall (0.92), and FPR (0.05).	4.5	Include
29	Feng (2025)	1.0: Utilizes CERT R4.2 (1,000 users, 70 anomalous). Conducts a pilot experiment (Sect 2) to quantify the imbalance (0.02% anomalies).	1.0: Features a hybrid of SMOTE oversampling for minority classes and random undersampling for normal behavior. Uses a chronological "extrapolation" partition (Sect 4.2).	1.0: Identifies specific "out-of-character" indicators across 5 logs: Logon, File, HTTP, Email, and Device. (Sect 4.2.1, Table 2).	1.0: Proposes MG-UABD , a two-stage system: Coarse-Grained Anomaly Detection (CAD) followed by Fine-Grained Anomaly Detection (FAD). (Sect 4.3, 4.4).	1.0: Reports 99.99% Precision , 99.99% Detection Rate (Recall) , and 99.99% F1-score on the behavior level. (Table 6).	5.0	Include
30	Ferraro (2025)	1.0: Uses the behavior-rich CERT r5.2 and the network-centric PicoDomain (Zeek logs). (Section 4.1)	1.0: Grouping logs by Activity ID (\$ID_k\$) and partitioning by category. Implements a re-prompting mechanism to ensure LLM outputs follow a strict JSON schema. (Section 3.1)	1.0: Specialized Agents are prompted to look for specific "red flags" like failed authentication (Kerberos errors) and unusual byte/packet counts. (Section 3.1, 3.3)	1.0: Proposes a Generative Agent-Based Modeling (GABM) framework using LLMs in a hierarchical multi-agent system (Specialized + Supervisor Agents). (Section 3)	1.0: Evaluates Accuracy, Recall, and FPR, emphasizing high recall (safety-first) and providing an ablation study. (Section 4, 5)	5.0	Include
31	Gayathri (2024) SPCAGAN	1.0: Uses the benchmark CERT r4.2 (1,000 users) and r5.2 (2,000 users) datasets covering 18 months of activity. (Sect. IV-A)	1.0: Derives a refined list of 47 features . Uses Min-Max normalization and strategic feature selection to remove multi-correlated/redundant variables. (Sect. III-A)	1.0: Incorporates sentiment analysis (abusive/unethical language) and psychometric details alongside traditional IT logs. (Sect. III-A)	1.0: Proposes SPCAGAN (PCA-based GAN) for data augmentation and a Hybrid Bayesian Deep Learning model for classification. (Sect. III-B, III-D)	1.0: Evaluates via Precision, Recall, F-score, and advanced metrics like Cohen's Kappa and Matthews Correlation Coefficient (MCC) . (Sect. IV-B)	5.0	Include
32	Gayathri R (2025) Adversarial Training	1.0: Utilizes CERT v4.2 and v5.2 (Section 4.1).	0.5: While the paper details the SHAP-based trigger generation (Algorithm 1), it fails to explicitly state the initial normalization method (e.g., Min-Max vs. Z-score) used for the raw tabular input features before they enter the SNN.	1.0: Specifically integrates Explainable AI (XAI) as an indicator of vulnerability and utilizes the Kerberos authentication error logs as a primary behavioral indicator (Section 3.3).	1.0: Proposes a highly detailed Self-Normalizing Neural Network (SNN) and 1DCNN architecture (Section 3.2).	1.0: Reports the Attack Success Rate (ASR) and the Performance Drop (\$M_{drop}\$) metrics across precision, recall, and F-score (Section 4.3).	4.5	Include

33	Gayathri B (2025) ITD-GMJN	1.0: Utilizes CERT , NSL-KDD , and UNSW-NB15 datasets for cross-environment validation. (Section 4, Page 123)	1.0: Detailed GOA-optimized MICE imputation for missing values and splitting into sequential/standalone logs. (Section 3, Page 119-120)	0.5: Identifies domain-based features for standalone data (e.g., first logon time) but lacks a deep rule-based logic catalog. (Section 3, Page 121)	1.0: Explains the LS-AE (LSTM-Autoencoder) for features and the Jordan Neural Network (JNN) for classification. (Section 3, Page 121-122)	1.0: Reports Accuracy (96%), Recall (92%), and computational timing (0.82s) compared against DBN, LSTM, and DNN. (Section 4, Table 3)	4.5	Include
34	Gonzales (2025)	0.5: Uses synthetic action sequences and UniProt (protein sequences) as a biological surrogate for user actions. While relevant, it lacks a dedicated cybersecurity benchmark like CERT or Enron. (Section III.A)	1.0: Detailed steps for Contiguous Pattern Mining , subpattern pruning , and n-gram generation . Standard label-encoding and padding are used for benchmarking. (Section II.A, III.C)	1.0: Grounded in the ABC (Antecedent-Behavior-Consequence) model from behavioral psychology to define the relationship between triggers and actions. (Section I.C)	1.0: Proposes the C3P and C3P-SMART pipelines, utilizing Kullback-Leibler (KL) divergence and Local Context Weighting (LCW) for scoring. (Section II)	1.0: Reports 90% Precision/Recall and 0.96 ROC-AUC . Compares performance against Isolation Forest, OC-SVM, and LOF. (Section III.B.4, Table 13)	4.5	Include
35	Greitzer (2021)	1.0: Utilizes the SOFIT Ontology combined with an expert survey of 20 professionals to create a 45-case synthetic testbed. (Sec 1, 2.2)	1.0: Details the Feature Engineering of raw "Level of Concern" ratings into Likelihood Ratios using a quadratic formula and antilog transformations. (Sec 4.2)	1.0: Defines a massive set of indicators (~200 factors) across five classes, specifically identifying behavioral "red flags." (Sec 2.2, Fig 2)	1.0: Implements and explains two distinct Bayesian Network structures: Indicator Bayes (Naïve) and Behavior Bayes (Hierarchical). (Sec 3.2, 5.1)	1.0: Provides clear comparative metrics ($\$R^2\$$) to show model improvement over standard counting and risk-summing methods. (Sec 5.2, Table 3)	5.0	Include
36	Rahman (2022)	1.0: Utilizes the CERT r4.2 benchmark dataset, consisting of 1,000 employees and 70 malicious users. (Section III.N)	0.5: Mentions pre-processing and "operational extraction" of attributes from logs but lacks specific mathematical normalization details. (Section III)	1.0: Defines a complex 7-dimensional conceptual model (Access, Motivation, Action, Intent, CAction, Method, Knowledge). (Section III.A-H)	1.0: Employs the NARX (Nonlinear Autoregressive Exogenous) neural network, providing full mathematical modeling. (Section III.O)	1.0: Reports $\$R^2\$$ (0.97), MAE, RMSE, and RAE across multiple data partitions. (Section IV, Tables 2-6)	4.5	Include
37	Haq (2022)	1.0: Employs the Enron e-mail dataset , a high-volume real-world dataset (517,401 emails). (Section 3, Fig 1)	1.0: Comprehensive Feature Engineering : email parsing (headers, body), date/time decomposition, and handling of missing values. (Section 4.1)	0.5: Uses the Person of Interest (POI) label as a ground truth indicator but focuses primarily on automated text analysis rather than expert behavioral rules. (Section 4.4)	1.0: Proposes hybrid LSTM models integrated with Word2vec and GloVe , compared against XGBoost , AdaBoost , and Random Forest . (Section 4.2-4.4)	1.0: Detailed performance metrics provided; XGBoost achieved the highest accuracy of 92% . (Section 5, Table 4)	4.5	Include
38	He (2022)	1.0: Utilizes multiple datasets: Enron (normal emails), monkey.org (phishing), and CMU-CERT r6.2 (insider	1.0: Proposes the MDFTS (Multi-Domain Feature Time Series) algorithm for heterogeneous data and uses Bi-LSTM for automatic feature	1.0: Incorporates the Big Five (OCEAN) personality model and psychological traits (curiosity, greed, anxiety) to analyze predisposition.	1.0: Detailed Double-Layer Framework : L-XGB (LSTM + XGBoost) for phishing and BiLA-ITD (Bi-LSTM + Attention) for insider threats. (Section: Proposed Scheme)	1.0: Comprehensive results: 98.35% accuracy for phishing and 96.4% precision for insider threats. (Section: Experimental Results, Tables 1-2)	5.0	Include

		threats). (Section: Dataset)	extraction without manual intervention. (Section: Overview)	(Section: Case Analysis, Table 3)				
39	He (2024)	1.0: Utilizes Enron , monkey.org , and CMU-CERT r6.2 . Covers 4,000 users over 18 months. (Sec IV.A)	1.0: Proposes the MDFTS algorithm for heterogeneous logs and uses Word Embedding (We) for behavior sequences. (Sec III.C)	1.0: Extracts features from 4 domains: behavior, text content, role (LDAP) , and psychology (OCEAN model) . (Sec III.C.2)	1.0: Detailed two-layer pipeline: L-XGB (LSTM+XGBoost) and BiLA-ITD (Bi-LSTM+Attention). (Sec III.B-C)	1.0: Exceptional results: 98.35% accuracy (Phishing) and 96.4% precision (Insider Threat). (Sec IV.E)	5.0	Include
40	Hong (2023)	1.0: Utilizes the CERT 4.2 dataset. Addresses the "imbalanced classification" challenge inherent in ITD. (Sec 1, 6)	1.0: Proposes a dual-stage feature engineering approach: manual selection + automatic extraction via LSTM Auto-encoder . (Sec 3.1)	1.0: Employs domain knowledge to select features from standalone activities and integrates them into a User-Day organizational graph . (Sec 3.2)	1.0: Introduces ResHybnet , a hybrid of GNN (topological) and CNN (non-topological) with a specific Residual Link . (Sec 3.3)	1.0: Demonstrates a 1.97% performance boost over state-of-the-art models and an F1-score improvement. (Sec 5)	5.0	Include
41	Huang (2025)	1.0: Introduces and utilizes specialized benchmark datasets: KDD-UEBA , NSL-UEBA , CIC-UEBA , and KDD-UNBLogs . (Section I, Section IV.C)	1.0: Defines ColumnTransformer encoding, one-hot labeling, Random Forest dimensionality reduction, and Batch Normalization scaling. (Section III.A-B)	1.0: Analyzes behavior from three expert perspectives: Temporal (periodicity), Sequential (order of operations), and Contextual (environment). (Section V.B.3)	1.0: Technically robust DenseAttDNN combining Dense Connections for feature reuse and Scaled Dot-Product Attention . (Section III.B)	1.0: Extensive evaluation using Accuracy (99.12%) , DR, FAR, and FNR with comparison to models like SEL and DeepFed. (Section V.B, Tables VII-X)	5.0	Include
42	Hurst (2022)	1.0: Utilizes pre-labelled real-world EHR logs collated from a UK-based hospital. (Abstract & Sec 3)	1.0: Clearly defines disaggregation into four groups and the use of 12 feature descriptors (Mean, Mode, SD, etc.). (Sec 3.1)	1.0: Identifies four distinct action categories: Routine, User, Patient, and Device interaction. (Sec 3, Fig 2)	1.0: Explains SVM, Random Forest, and Decision Tree. Provides specific SVM parameters (C-classification, Radial Kernel). (Sec 3.2 & 4.2)	1.0: Provides Accuracy (0.9908), Kappa, Sensitivity, and Specificity with comparative analysis across three models. (Sec 4.3, Table 10)	5.0	Include
43	Jaiswal (2024)	1.0: Employs CERT r4.2 and r5.2 , the most recognized benchmark datasets for ITD. (Sec 3.1)	1.0: Clearly defines activity encoding, day-wise grouping, outer joins, and Min-max scaling . (Sec 3.1, Algorithm 1)	0.5: Identifies 30 attributes based on activity frequency. While useful, it lacks the complex behavioral or psychological rules found in expert-driven models. (Sec 3.1, Table 2)	1.0: The TSITD (Two-Step Insider Threat Detection) pipeline is technically sound, exploring 40 combinations of sampling and classifiers. (Sec 3.2)	1.0: Uses robust metrics including Balanced Accuracy (BAcc) and the TOPSIS method for model ranking. (Sec 4.2, Table 6)	4.5	Include
44	Janjua (2021)	1.0: Employs the Enron dataset (0.5M emails) and the TWOS (Wolf of SUTD) dataset. (Sec 3.1)	1.0: Details missing value imputation, stop word removal, stemming, and TF-IDF vectorization. (Sec 3.3–3.4)	0.5: Identifies metadata (sender, header) and mentions psychological personality assessments in TWOS, but focuses primarily on textual clustering. (Sec 3.1.2)	1.0: Detailed semi-supervised pipeline: K-Means for labeling followed by Decision Tree for prediction. (Sec 3, Algorithm 1)	1.0: Reports 99.96% accuracy and 0.994 AUC for the Decision Tree, with comparative analysis. (Sec 6)	4.5	Include

45	Jyosthna (2021)	1.0: Utilizes the CERT r6.1 benchmark dataset (4000 users). (Sec I & III)	1.0: Detailed steps for Min-Max normalization , statistical aggregation of log events, and PCA for dimensionality reduction. (Sec III.E-F)	0.5: Employs LDAP roles and frequency counts of activities (login/logoff, device connectivity), but lacks deep behavioral or psychological feature engineering. (Sec III.B)	1.0: Technically sound unsupervised method (N-Median Outlier Detection - NMOD) clearly compared against DBSCAN and k-NN. (Sec III.H.3)	0.0: Findings lack standard performance metrics (AUC, F1, Precision/Recall). Evaluation is restricted to qualitative lists of detected outliers. (Sec IV)	3.5	Include
46	Kabir (2023)	1.0: Utilizes the CERT 4.2 benchmark dataset, incorporating both threat and non-threat baseline records. (Sec 3)	1.0: Detailed steps for Unix epoch conversion , text-to-integer encoding, and manual label mapping (HTTP=0, Email=1, etc.). (Sec 3)	0.5: Identifies basic log attributes (Date, User, PC, Activity) but lacks deep behavioral or psychological rule-based logic. (Sec 3, Fig 1)	1.0: Well-explained Keras sequential model with three dense layers, ReLU activation, and Dropout for regularization. (Sec 4)	1.0: Reports 100% Accuracy, Precision, Recall, and F1 across 75% and 80% training splits. (Sec 5, Table 1)	4.5	Include
47	Kamatchi (2025)	1.0: Employs a simulated dataset via NS-3.26 and the benchmark X-IIoTID dataset. (Sec 4.1)	1.0: Details mean imputation for missing values and scaling of login times/transmission metrics. (Sec 3.3)	0.5: Identifies access patterns, login times, and transmission metrics. These are standard features but lack deep psychological profiling logic. (Sec 3.3)	1.0: Robust Federated Learning (FL) architecture using Bi-LSTM local models and FAWO aggregation. (Sec 3.3-3.4)	1.0: Reports accuracy (98.85% simulated), throughput, and uses t-tests to prove statistical significance. (Sec 4.5.2-4.5.3)	4.5	Include
48	Kan (2023)	1.0: Utilizes the CERT 4.2 dataset containing 1,000 user activity records over 501 days. (Sec 2.1)	1.0: Extremely detailed Data Adjusting (DA) strategy involving misclassification data retention, clustering under-sampling, and SMOTE. (Sec 3.2, Algorithm 1)	0.5: Identifies 40 features based on daily activity frequency . While it segments data into "Work hours" and "Off hours," it relies primarily on counts rather than complex sociotechnical rules. (Sec 2.1, Fig 1)	1.0: Proposes ERPSO-XGBoost , a hybrid model using an Ethnic Randomized Particle Swarm Optimization to tune XGBoost hyperparameters. (Sec 4)	1.0: Performance is validated via Precision, Recall, F1-score, and AUC against multiple state-of-the-art baselines. (Conclusion)	4.5	Include
	Kandanaarachchi (2021)	0.5: Uses data from the "LAN Security Monitoring Project." While real-world, it is not a specialized ITD benchmark like CERT or Enron.	0.5: Focuses on network protocol packets (ARP, TCP, UDP) rather than organizational user-log sequences.	0.0: Zero expert-defined behavioral or psychological indicators for human insiders are utilized.	0.5: The "Honeyboost" framework is designed for honeypot enhancement and general Network Anomaly Detection (NADS), not specifically for insider behavior.	0.5: Detects "anomalous nodes," but lacks ground-truth verification of actual insider maliciousness (e.g., sabotage, theft).	2.0	Exclude
49	Kim (2023)	1.0: Employs the benchmark CERT dataset (Scenarios 1, 2, and 3). (Sec 3.1)	1.0: Detailed preprocessing including scaling, statistical frequency analysis, and Discrete Wavelet Transformation (DWT) for denoising. (Sec 3 & 4)	0.5: Focuses on standard log-based features (Logon/Logoff, device, email, http, file) without deep integration of psychological red flags. (Sec 3.1, Table 1)	1.0: Technically sound use of OC-SVM optimized via DWT; compares seven different wavelet functions (Haar, Daubechies, etc.). (Sec 3.2-3.3)	1.0: Provides comprehensive metrics: Precision, Recall, F1-score, AUC , and False-Positive Rate (FPR) . (Sec 4, Table 4)	4.5	Include

50	Kong (2025)	1.0: Validated on four benchmark datasets: CERT r4.2, SEA, HCT, and Purdue University UNIX (PU) . (Sec V.A)	1.0: Comprehensive behavior mapping using a specific formula ($\text{Action ID} = \text{Factor} \times 8 + \text{Offset}$) and a novel tagging strategy based on contextual scores . (Sec IV.E.1-2)	1.0: Proposes highly sophisticated expert features: UADS (User Action Diversity Score) using entropy and BCS (Behavioral Context Score) for global/local importance. (Sec IV.D)	1.0: Advanced DPI-ITD framework featuring a dual-perspective modeling module and Adaptive GloVe embedding. (Sec IV.A, IV.E.3)	1.0: High-quality results provided for F1-score (up to 96.59%), FPR, and ablation studies across all datasets. (Sec V.D-E)	5.0	Include
51	Kotb (2025)	1.0: Uses the complete CERT r4.2 dataset, including all three malicious scenarios and 1000 users. (Sec 3.1)	1.0: Employs Deep Feature Synthesis (DFS) to automatically convert relational log files into structured tabular user profiles. (Sec 3.2)	0.5: Incorporates the "Big Five" personality scores (psychometric entity) but relies primarily on automated mathematical aggregations (Relational features). (Sec 3.2)	1.0: Innovative DS-IID model combining DFS with an 11-block binary deep learning architecture and generative AI (GANs/VAEs) for adversarial testing. (Sec 3.3)	1.0: Reports 97.31% accuracy and 0.99 AUC, validated across 9 distinct performance metrics. (Sec 5)	4.5	Include
52	Lavanya (2024)	1.0: Employs the benchmark CERT r6.1 and r6.2 datasets, and IoT enabled institutional log records involving 3,995 users and 5 scenarios. (Sec IV.A.1)	1.0: Comprehensive pipeline including Improved PCA (IPCA) for functionality extraction and outlier-resistant k-means for scenario clustering. (Sec IV.A.1)	1.0: Maps CERT scenarios (S1-S5) directly to IoT architecture layers (Perception, Transport, etc.), providing deep contextual grounding. (Sec IV.A.1, Table 4)	1.0: Proposes a novel Enhanced BiGAN (EBiGAN) for data augmentation and a DNN-PI (Deep Neural Network with Bayesian Optimization). (Sec IV.A.2-3)	1.0: Reports high metrics: Accuracy (98.8%), Precision (97.8%), and Detection Rate (96.7%) with low false alarm rates. (Sec V.E, Table 6)	5.0	Include
53	Le and Zinc-Heywood (2021) Exploring anomalous behaviour	1.0: Utilizes multiple benchmark releases: CERT r4.2 and r5.2 for training, and r4.1, r5.1, and r6.2 for unseen testing.	1.0: Extremely detailed preprocessing that reasons for auxiliary info (PC ownership, work hours) and evaluates three levels of temporal granularity (session, day, week).	0.5: Identifies behavioral indicators like USB connection lengths, role/team context, and HTTP job categories, but lacks deep psychological modeling.	1.0: Proposes a hybrid unsupervised-supervised pipeline that incorporates an analyst-in-the-loop "Investigation Budget" and compares multiple feature engineering methods (AE, PCA, RP).	1.0: Validated with standard metrics at both the data-instance level and the user-based level across five distinct CERT releases.	4.5	Include
54	Le and Zinc-Heywood (2021) Unsupervised ensembles	1.0: Uses CERT r4.2, r6.2, LANL, and TWOS . High diversity in data sources. (Sec IV.A)	1.0: Innovative Percentile Representation ($\$P_w\$$) and Concatenation to encode temporal shifts. (Sec III.B)	0.5: Focuses on "behavioral changes" (e.g., job-seeking, late-night logins) but relies on log frequency rather than psychological profiling. (Sec V.A)	1.0: Proposes an Unsupervised Ensemble combining Autoencoders, Isolation Forest, LODA, and LOF using a voting scheme. (Sec III.C-D)	1.0: Extensive sensitivity analysis on training duration, poisoning, and user counts. Strong AUC/UAUC results. (Sec IV.C-E)	4.5	Include
55	Li Dongyang (2021) Image-Based	1.0: Employs CERT r4.2 (1,000 users) for main experiments and r6.2 (4,000 users) for robustness validation. (Sec 4.1)	1.0: Innovative Image Conversion strategy that transforms 1D feature vectors into 2D grayscale images via an outer product ($\$s = x^T \cdot x\$$). (Sec 3.3)	1.0: Feature set specifically designed for malicious scenarios, including work hours, PC ownership, and specific file/web categories. (Sec 3.2, Table 1)	1.0: Proposes IGT , a novel self-supervised framework that converts anomaly detection into a supervised image classification task using a Wide Residual Network (WRN-10-4) . (Sec 3.4)	1.0: Reports 92.13% user-based AUROC, outperforming traditional Autoencoder and Isolation Forest baselines by significant margins. (Sec 4.4, Table 6)	5.0	Include

56	Li Ximing (2023) Dual-Domain Graph	1.0: Uses CERT r4.2 (1,000 users) and r6.2 (4,000 users). Comprehensive multi-scenario evaluation. (Sec IV.A.1)	1.0: Employs Natural Language Processing (NLP) for log content and a Weighted Feature Similarity Mechanism to transform logs into heterogeneous graphs. (Sec III.B)	1.0: Uses domain-specific features (Wikileaks/hacking web visits, off-hour logins, removable device connections) to drive node attributes. (Sec III.B, Fig 6)	1.0: Proposes a novel Dual-Domain Graph Convolutional Network (DD-GCN) that integrates topology and feature-similarity domains using an attention mechanism. (Sec III.A)	1.0: Achieves 98.65% accuracy, significantly outperforming SOTA (RF, SVM, LSTM, CNN, GCN). Validated with PR-curves and ablation studies. (Sec IV.C, Fig 8)	5.0	Include
57	Li Ximing GMFITD: Graph Meta-Learning (2024)	1.0: Utilizes CERT r4.2 , r6.2 , and the UMDWikipedia dataset. Addresses data scarcity in real-world scenarios. (Sec V.A.1)	1.0: Proposes a Structural Reconstruction Mechanism using Graph Auto-Encoders (GAE) to establish "soft" behavioral relationships. (Sec IV.A)	1.0: Differentiates between "hard" (direct communication) and "soft" (behavioral similarity) relationships. Uses specific log attributes for node features. (Sec I.A, V.A.1)	1.0: Proposes GMFITD , a sophisticated meta-learning framework integrating GAE, Prototypical GNNs, and Model-Agnostic Meta-Learning (MAML) . (Sec IV.C-D)	1.0: Achieves ~87% accuracy in 5-shot scenarios and demonstrates significant resilience against targeted/non-targeted adversarial attacks. (Sec V.C)	5.0	Include
58	Lilhore et al. (2025)	0.5: Does not use a dedicated ITD dataset . Relies on network-centric NIDS benchmarks (CIC-IoT/UNSW) which lack human behavior.	1.0: Technical steps (SMOTE, Z-score, Min-Max) are explicitly defined. (Section: Pre-processing)	0.5: Identifies "expert" features, but they are network-flow metrics (packet rate) rather than human indicators (keystrokes, sentiment).	1.0: The CNN-LSTM-Transformer + RL hybrid is technically robust and clearly diagrammed.	1.0: Provides high-resolution metrics and comparative analysis against other hybrids.	4.0	Include
59	Liu et al. (2025)	1.0: Uses the gold-standard CERT-r4.2 and CERT-r5.2 datasets. And DARPA 1999 (Section 5.1)	1.0: Detailed session-based segmentation (Logon-Logoff units) and BERT-based vocabulary representation. (Section 3.3.1)	1.0: Clearly defines "motive" vs "behavioral" evidence, including job-site browsing and night-time uploads. (Section 3.2)	1.0: Technically sound integration of BERT, Knowledge Graphs, and Preference Propagation. (Section 3)	1.0: High F1-scores (0.96–0.99) with ROC curves and comparison against 6 baseline models. (Section 5.1/5.2)	5.0	Include
	Mahmoudi-Nasr (2024)	0.0: Uses a private simulation (UMZ ICS Lab) based on Iranian power system logs. No public ITD benchmark (CERT/Enron).	0.5: Describes CIM mapping, but lacks standard ML preprocessing like scaling or encoding for model training.	1.0: Identifies specific grid-based indicators (pending/cleared alarms). (Section 4)	0.0: Uses manual formulas (EWMA) and statistical control charts (I-MR). Does not use ML/DL architectures.	0.5: Results focus on access authorization success rather than standard ITD metrics like AUC or F1-score.	2.0	Exclude
60	Manoharan et al. (2024)	1.0: Uses the benchmarked CERT r4.2 dataset. (Section 3.1)	1.0: Detailed steps for merging logs, mean imputation, categorical encoding, and downsampling. (Section 3.1, 3.2)	1.0: Identifies specific features like Role, Dept, and activity codes (1-7) for user-day granularity. (Table 2)	1.0: Evaluates 8 technically sound supervised ML models (RF, XGB, MLP, etc.) with specific parameters. (Table 3)	1.0: Reports Accuracy, F1, Precision, and Recall; includes comparative analysis with 4 existing works. (Table 11)	5.0	Include
61	Matos et al. (2025)	0.5: Uses a synthetic healthcare access log generator (50,500 events). Not a standard benchmark like CERT, but	1.0: Defines stratified 80/20 splits, Autoencoder reconstruction error scaling, and categorical encoding for	1.0: Lists 5 specific behavioral rule violations (e.g., off-hours, cross-dept, wrong patient pool). (Section 3.2)	1.0: Features a complex hybrid Meta-classifier fusing Autoencoder, LightGBM, and expert rules. (Section 3.3)	1.0: Reports F1 (0.89), Recall (0.81), and Precision (0.99) compared against 7+ baselines including	4.5	Include

		justified for HIPAA privacy. (Section 3.2)	roles/departments. (Section 3.3/3.4)			LSTM and GNN. (Table 2)		
62	Medvedev et al. (2025)	0.5: Utilizes CMU and KeyRecs datasets. (Section 4.1)	1.0: Detailed interpolation for variable password lengths and GAFMAT transformation into 40x40 images. (Section 3.1)	1.0: Extracts specific timing features: hold times, keydown-keydown, and keyup-keydown intervals. (Section 4.1)	1.0: Employs Siamese Neural Networks (SNN) with triplet loss and CNN branches. (Section 3)	1.0: Reports Equal Error Rate (EER) of 0.11–0.12 and provides comparative discussion. (Section 6)	4.5	Include
63	Mehmood et al. (2023)	1.0: Uses a customized dataset derived from the benchmarked CERT dataset. (Section I/IV)	0.5: Outlines basic outlier removal (averaging) and imputation, but lacks a detailed deep-dive into the specific normalization formulas or sequence-handling depth. (Section III.E)	0.5: Identifies common behavioral scenarios like off-hours activity, but does not define a comprehensive set of complex behavioral rules or psychosocial indicators. (Section II)	1.0: Implements and explains 4 ensemble models (RF, AdaBoost, XGBoost, LightGBM). (Section III)	1.0: Provides Accuracy, Precision, Recall, F1, and False Alarm Rates with comparative analysis. (Section IV.2)	4.0	Include
64	Mehnaz & Bertino (2021)	0.5: Uses Wikipedia repository logs (77 users). While real-world, it is not a standard ITD benchmark like CERT/Enron. (Section 6.1)	1.0: Uses blktrace for block-level I/O capture; detailed online sector-to-inode translation. (Section 2.1, 4.1)	0.5: Identifies block-level access size, relative file segments, and multi-level temporal structures. (Section 4.2, 4.4.2)	1.0: Technically robust framework using Serial/Parallel Episode Mining and Finite State Automata. (Section 4.3)	1.0: Reports Accuracy (98.7%), Precision, Recall, and F-measure; includes a comparison of 6 taxonomy cases. (Section 6.4)	4.0	Include
65	Mladenovic et al. (2024)	1.0: Evaluated on the benchmarked CERT r4.2 dataset across six distinct experiments. (Section: Abstract)	1.0: Employs TF-IDF for text vectorization and SHAP for feature interpretability. (Section: TF-IDF / SHAP)	1.0: Focuses on "Sentiment and Context" of actions, identifying shifts in tone and keyword significance. (Section: Conclusion)	1.0: Implements XGBoost and AdaBoost optimized by a novel Modified Red Fox Optimization (MRFO) algorithm. (Section: Methods)	1.0: Reports Accuracy, Precision, Recall, and F1; uses Shapiro-Wilk tests for statistical significance. (Section: Statistical Validation)	5.0	Include
66	Nasir et al. (2021)	1.0: Uses the benchmarked CMU CERT synthetic dataset r4.2. (Section III.A)	0.5: Describes aggregating multiple CSVs, mean imputation for missing values, and integer encoding. Standard preprocessing approach. (Section III.C)	0.5: Utilizes standard organizational indicators such as Logon/Logoff, USB usage, and user job roles. (Section III.D)	1.0: Proposes a technically sound LSTM-Autoencoder architecture specifically for multivariate time-series data. (Section III)	1.0: Reports Accuracy (90.60%), Precision (97%), and F1 Score (94%) with a comparison to several DL baselines. (Section V.A)	4.0	Include
67	Nikiforova et al. (2024)	0.5: Uses real-world audit logs from 488 users (2.7M actions) in a business environment. While realistic, it is a proprietary dataset rather than a public benchmark like CERT.	1.0: Detailed steps for anonymization, mapping source-to-target data, and creating a matrix of user-action pairs for Markov transition probabilities.	1.0: Identifies specific behavior indicators through transition probabilities between action pairs (e.g., login → view_menu) and session anomaly coefficients.	1.0: Implements a two-level automated approach using K-means clustering (with the Elbow method for optimal \$K\$) and Markov chain behavioral modeling.	1.0: Provides quantitative results for session identification, graphical representations of suspicious sessions, and identifies anomalies for 158 users.	4.5	Include

68	Pal et al. (2023)	1.0: Utilizes three versions of the benchmarked CMU CERT dataset (v4.2, v5.2, v6.2) for cross-version validation. (Section 3.1)	1.0: Detailed preprocessing including "Equally-Weighted Random Sampling" (EWRS) to handle scenario-specific imbalance and one-hot encoding of 282 action-IDs. (Section 3.1.2)	1.0: Develops a massive set of 282 action-IDs that distinguish between assigned vs. supervisor PCs, and hidden (BCC) vs. non-hidden emails. (Table 4/Section 3.1.2)	1.0: Proposes a complex ensemble of stacked-LSTM and stacked-GRU models with integrated attention layers. (Section 3.2.1)	1.0: Reports a near-perfect AUC (0.99) and uses an ablation study to prove the impact of each model component. (Section 4.5.3)	5.0	Include
69	Patel & Iyer (2025)	1.0: Utilizes the benchmarked Landauer et al. (2022) UEBA (User and Entity Behavior Analytics) dataset for cloud computing.	0.5: some feature extraction details (admin, auth, comments) and binary data encoding to lay the foundation for modeling.	1.0: Employs the FP-Growth algorithm to identify recurring sequences and set threshold-based frequent mining parameters.	1.0: Proposes a novel hybrid of Siamese Convolutional Neural Networks (SCNN) and Deep Neural Networks (DNN) for similarity learning.	1.0: Reports Accuracy (91.1%), FPR (7.3%), and TPR (90.7%).	4.5	Include
70	Peccatiello et al. (2023)	1.0: Uses the benchmarked CMU CERT v4.2 dataset. (Section IV.A)	1.0: Implements sessionization (login to logoff), implicit logoff insertion, and specific feature extraction (e.g., TF-IDF for job sites). (Section IV.B)	0.5: Utilizes standard behavioral indicators found in the CERT documentation (login times, USB counts) without introducing novel behavioral or psychological rules. (Section IV.B)	1.0: Proposes a framework for stream-based detection using one-class algorithms and a periodic retraining mechanism to handle concept drift. (Section III.A)	1.0: Provides Precision, Recall, F1, Kappa, and Gmean metrics, with a focus on both malignant and benign class performance. (Section V)	4.5	Include
71	Pérez-Gomariz et al. (2025)	0.5: Validated on a real-world cybersecurity incident at a Fortune 500 company. While highly realistic, it is a proprietary dataset rather than a public benchmark. (Section 1)	0.5: Focuses on Windows Event ID 4624; builds directed graphs and extracts user activity sub-graphs via a 24-hour adjacency matrix algorithm. (Section 5.2.1)	0.5: targets APT tactics like Living off the Land (LotL), Pass-the-Hash, and RDP/PowerShell/WMI-based movement. (Table 1)	1.0: Proposes an unsupervised seq2seq (Transformer) architecture to learn network dynamics and machine roles. (Section 3.2)	1.0: Successfully identified 86 malicious movements out of 15,770 in a live network of thousands of nodes. (Section 5.2.2)	3.5	Include
72	Prasad et al. (2024) Enhanced insider threat	1.0: Utilizes the benchmarked CERT dataset (v5.2). (Section 3.1)	1.0: Comprehensive preprocessing including missing value/outlier checks, six different resampling methods (Oversampling: Random, SMOTE, ADASYN; Undersampling: Random, Cluster Centroids, ENN), and PCA for dimensionality reduction. (Section 3.2, 3.3)	0.5: Uses standard behavioral indicators derived from the CERT dataset logs. (Section 3.1)	1.0: Proposes a multi-stage automated framework using five ML base-learners and two ensemble techniques (Stacking and Voting Classifiers). (Section 3.3)	1.0: Reports a high accuracy of 99% and compares results against several existing models (SVM, XGBoost, Isolation Forest). (Section 4.2, 4.3)	4.5	Include
73	Prasad et al. (2025) Hybrid	1.0: Utilizes the benchmarked CERT	1.0: Extensive pipeline: StandardScaler, IQR for	1.0: Defines highly specific behavioral	1.0: Proposes a novel hybrid architecture fusing generative	1.0: Reports Accuracy (93.4%), Precision	5.0	Include

	Machine Learning	dataset (~693k samples) with a documented imbalance of 1:100 to 1:500. (Section 3/4.1)	outliers, and a comparison of three resampling techniques (SMOTE, Undersampling, ENN). (Section 3.2)	features including "logins between 9 PM and 6 AM," "sensitive directory access," and "external email recipient ratios." (Section 3.1.1)	latent features (from DAE/VAE) with discriminative classifiers (RF/XGBoost). (Section 3.3)	(91.2%), Recall (90.5%), and AUC-ROC (0.95); includes statistical significance testing (t-test). (Section 5.1/5.2)		
74	Prasad et al. (2025) Deep Learning Techniques	1.0: Utilizes the standard CMU CERT benchmark dataset with 830 features and ~694k entries.	1.0: Implements SMOTE for class imbalance and a novel 8-way behavioral feature categorization strategy.	1.0: Identifies and organizes 830 features into expert-defined behavioral groups (Time, User, Project, Activity, Logon, USB, File, Email).	1.0: Evaluates a comprehensive suite of ML classifiers (RF, DT, KNN, Adaboost) alongside Artificial Neural Networks (ANN).	1.0: Provides exhaustive accuracy, precision, and recall metrics for every classifier across all eight behavioral categories.	5.0	Include
75	Qawasmeh & AlQahtani (2025)	0.5: Utilizes a synthetic dataset (10,000 records, 500 users) designed to mimic real-world scenarios rather than a standard public benchmark. (Sec 4.1)	1.0: Clearly defines SMOTEENN for imbalance, missing value imputation (impute with 1), and outlier clamping. (Sec 4.3, Fig 3)	1.0: Identifies 17 specific anomalous behavior types (e.g., USB insertion, secure printing) with associated severity weights (4-10). (Table 2)	1.0: Technically sound real-time monitoring and classification workflow using RF, LR, XGBoost, and SVM. (Sec 3.1, Fig 1)	1.0: Provides exhaustive metrics (Accuracy, Precision, Recall, F1) and a massive comparative analysis with 30+ studies. (Sec 5.4, Table 5)	4.5	Include
76	Qi et al. (2025)	1.0: Uses the benchmarked CERT r4.2 and LANL (Los Alamos National Laboratory) datasets. (Sec 5.2)	1.0: Detailed optimization including domain name mapping, USB/File aggregation per PC node, and subgraph sampling for imbalance. (Sec 4.1, 5.3)	1.0: Defines "Heterogeneous Degree" and extracts specific spatial, temporal, and attribute features. (Sec 4.1.3, Table 2)	1.0: Technically sound ATHITD model combining attention mechanisms for spatial/short-term temporal encoding and Transformers for long-term evolution. (Sec 4)	1.0: Reports F1 scores of 0.96 (CERT) and 0.97 (LANL) with clear comparative analysis. (Abstract, Sec 5.2)	5.0	Include
77	Racherache et al. (2023)	1.0: Uses two real-world network traffic datasets (Small and Large) provided by industry partners. (Sec 5.2)	1.0: Detailed steps including feature filtering, duration calculation, MAC vendor lookup, SMOTE up-sampling, and Min-Max scaling. (Sec 3.1, 3.2)	1.0: Defines "Cyber-Persona" types and lists 20+ features like Application_category and hostname embeddings. (Sec 2.2, Table 1)	1.0: Comprehensive multi-module framework using CNN, LSTM, Random Forest, and Isolation Forest for various stages. (Sec 4)	1.0: Reports high performance metrics across all modules (F1-score of 98.89% and AUC of 99.95%). (Abstract, Sec 1)	5.0	Include
78	Randive et al. (2023 a) pattern-based approach	1.0: Specifically utilizes the CMU CERT r4.2 benchmark dataset spanning 500 days. (Section 4.1)	1.0: Clearly defines a pipeline including 1D feature vector construction, SMOTEENN for imbalance, and conversion to grayscale images. (Section 3.2, 4.1)	0.5: Identifies statistical and frequency-based features but adopts the specific set from Chattopadhyay et al. [27] rather than defining novel behavioral logic. (Section 4.3)	1.0: Proposes a Wavelet CNN (WCNN) that integrates multiresolution analysis (wavelets) into the CNN paradigm to capture spectral features. (Section 3.1)	1.0: Reports 97.19% accuracy and 97.30% AUC with comparisons to MobileNet V2, ResNet-50, and VGG-19. (Section 5, 6)	4.5	Include
79	Randive (2023b) MWCapsNet	1.0: Utilizes both CERT r4.2 and r5.2 benchmark datasets. (Sec 4.1)	1.0: Details a sophisticated pipeline using IGTD (Image Generator for Tabular Data) based on Pearson	0.5: While it describes four detailed scenarios (S1–S4), it acknowledges that the feature vectors are	1.0: Proposes the MWCapsNet , a technically advanced model integrating Wavelet decomposition into a Capsule Network with dynamic routing. (Sec 3.3, 4.5)	1.0: Reports 98.88% accuracy and 99.21% precision with low False Positive Rates. (Abstract, Conclusion)	4.5	Include

			correlation and Manhattan distance. (Sec 4.4)	adopted from existing work [10, 24, 37] rather than being entirely novel expert definitions. (Sec 4.3)				
80	Rauf et al. (2021)	1.0: Evaluated on a real-life threat test dataset (CERT v4.2). (Abstract, Section 5)	1.0: Defines temporal criteria to convert unstructured logs into structured "activity vectors" and defines a mathematical weighted distance metric. (Section 1.1)	1.0: Maps biological sensing principles to specific system activities and mathematically formalizes behavioral deviations. (Section 4, Fig 3)	1.0: Proposes a dual-layered bio-inspired framework: Threat Analytics Module (TAM) and Policy Regulation Module (PRM) using formal state-transition logic. (Section 4)	1.0: Uses formal methods (Z3 Theorem Prover) to verify policy correctness and evaluates threat detection via distance-based ML. (Section 4.4, 5)	5.0	Include
81	Saminathan et al. (2023)	0.5: Utilizes a simulated organizational environment and a "real-life threat test dataset" collected via custom logging scripts. (Sec 3.1, 4.1)	1.0: Details removal of null values and Min-Max normalization scaling features to a [0, 1] range. (Sec 3.4)	1.0: Defines highly specific indicators including keystroke dynamics, mouse coordinates across 9 quadrants, and granular resource usage. (Sec 3.2.1, 3.2.2)	1.0: Clearly explains an 8-layer symmetric ANN-Autoencoder architecture (82-64-32-16-8-16-32-64-82) with ReLU activation. (Sec 3.5, Fig 6)	1.0: Reports 94.3% accuracy, 11.1% FPR, and 89.1% precision, with comparisons to existing UBA methods. (Abstract, Sec 6)	4.5	Include
82	Bin Sarhan & Altwaijry (2023)	1.0: Utilizes the benchmarked CERT r4.2 dataset, documenting 1000 employees and specific malicious scenarios. (Section 3.1)	1.0: Comprehensive pipeline including One-Hot encoding, Deep Feature Synthesis (DFS), PCA (95% variance), and SMOTE. (Section 3.3)	1.0: Identifies specific manual features (e.g., Weekday_Logon_After) and maps behaviors like thumb drive usage. (Table 4, Section 3.2)	1.0: Evaluates multiple models (SVM, NN, AdaBoost, iForest) with detailed hyperparameter tuning via Grid-Search. (Section 3.4)	1.0: Reports Accuracy, Precision, Recall, and F1-Score with 10-fold cross-validation and a comparative table. (Section 4, Table 8)	5.0	Include
83	Senevirathna et al. (2025)	1.0: Utilizes the benchmarked CERT Insider Threat Identification Dataset for cybersecurity and domain-specific violent/non-violent video clips for physical security. (Sec III-A & III-B)	1.0: Defines a rigorous pipeline including mean/median imputation, backward/forward filling for logs, and spatial/pixel normalization (224x224) for video frames. (Sec III-A & III-B)	1.0: Maps behavioral indicators to the OCEAN psychometric model, logon frequency, and specific device/file interaction patterns. (Sec III-A)	1.0: Technically sound dual-layered architecture: CNN-Random Forest Ensemble for behavioral logs and Hybrid CNN-LSTM (MobileNetV2 backbone) for video surveillance. (Sec III-A & III-B)	1.0: Provides exhaustive metrics including Precision, Recall, and F1-score across benign/malicious classes, achieving up to 98% accuracy. (Sec IV-A, Table III)	5.0	Include
84	Singh et al. (2022) Multi Fuzzy Classifier	1.0: Uses the benchmarked CMU-CERT v4.2 dataset, which includes 1000 users over 18 months. (Sec 1, Table 1)	1.0: Detailed steps for data cleaning, decimal scale and z-score normalization, and redundancy reduction via high correlation filters. (Sec 3.4)	1.0: Identifies specific time-dependent, frequency-dependent, and content-based behavioral features, including OCEAN psychometric traits. (Sec 3.5)	1.0: Proposes a technically sound Multi Fuzzy Classifier (MFC) using parallelized Mamdani inference engines. (Sec 3.8)	1.0: Evaluates across eight metrics (Accuracy, Precision, Recall, F-measure, etc.) and compares against four state-of-the-art methods. (Sec 4.2, Table 8)	5.0	Include
85	Singh et al. (2023)	1.0: Utilizes the benchmarked CMU-	1.0: Clearly defines missing value imputation	1.0: Identifies time-based behavioral	1.0: Proposes a sophisticated hybrid of Bi-LSTM, ANN-based	1.0: Supported by Accuracy, Precision,	5.0	Include

	Hybrid Learning Approach	CERT r4.2 dataset. (Section 4)	and normalization (Decimal and Z-score). (Section 3.3)	features and the concept of "behavioral drift" to flag risky actions. (Section 3.5)	selection, and SVM classification. (Section 3.1)	Recall, and F-measure, with comparative analysis against four state-of-the-art methods. (Section 4, Page 4592)		
86	Sivakrishna et al. (2025)	1.0: Utilizes the benchmarked CMU CERT r4.2 for training and generalizes testing on r5.2 and r6.2. (Section 3.1)	1.0: Clearly defines daily aggregation, missing value imputation, and a hybrid SMOTEENN balancing technique. (Section 3.2, 3.3)	1.0: Proposes 25 features, including 5 novel engineered behavioral temporal features (FAR, EDS, IAV, OAF, WF). (Section 3.2.4)	1.0: Employs XGBoost optimized via Bayesian Optimization , with a formal mathematical breakdown of the objective function and surrogate models. (Section 3.4)	1.0: Reports Accuracy (99.0%), Precision, Recall, F1, and ROC-AUC (99.7%) with cross-version validation and comparative analysis. (Section 4, Table 12)	5.0	Include
87	Song et al. (2024)	1.0: Utilizes the benchmarked CMU CERT v6.2 dataset, which simulates 4,000 employees over 516 days.	1.0: Employs a unique "adaptive" extraction method that selects the best time granularity (G) and starting bias (b) for each user based on maximizing covariance.	1.0: Focuses on "behavioral rhythm," distinguishing between absolute time (24h position) and relative time (sequence order).	0.5: Relies on a standard Stacked BiLSTM (3 layers) and FNN; the architectural novelty is primarily in the input encoding rather than a new paradigm (e.g., Bayesian optimization or parallelized logic).	0.5: Reports strong metrics (AUC 0.9730) but lacks cross-version or cross-dataset validation, and explicitly notes the "discrepancy" between its synthetic data and real-world distributions.	4.0	Include
	Sulaiman et al. (2023)	0.0: Does not use behavioral logs (e.g., CERT r4.2). It uses numerical scenarios to solve Ordinary Differential Equation.	0.0: No behavior-based preprocessing; focuses on mathematical state initialization.	0.0: No behavioral indicators; uses epidemic parameters like infection and patch removal rates.	0.0: Uses ML (ANN) as a "surrogate solver" for math, not as a detection/classification framework.	0.5: Reports MSE and convergence for math solutions, not detection metrics like F1 or Precision for identifying users.	0.5	Exclude
88	Tabassum et al. (2024)	1.0: Uses a massive real-world dataset of 1,007,727 EHR audit log entries from a North England hospital. (Dataset Section)	1.0: Rigorous preprocessing including mean/mode imputation, One-hot encoding, and Min-Max normalization. (Preprocessing Phase)	1.0: Identifies domain-specific indicators: "access beyond discharge date," "unusual duration spent on records," and "increased action frequency." (Conclusion)	1.0: Proposes a hybrid unsupervised-supervised pipeline using Isolation Forest (IF) and LOF combined with SVM/Decision Trees. (Methodology)	1.0: Reports Accuracy (99.21%), Sensitivity, Specificity, and F1-score, validated via Silhouette and Dunn scores. (Results)	5.0	Include
89	Tao et al. (2023) Temporal convolutional network	1.0: Uses the SapiMouse dataset (120 subjects), a standard benchmark for behavioral mouse dynamics. (Section 5.2)	1.0: Detailed preprocessing including "action segmentation," first-order difference (dx, dy), smoothing, and normalization. (Section 5.3)	1.0: Identifies key temporal and spatial indicators: XY coordinates, event types (drag, click), and timestamps. (Section 4.2.1)	1.0: Proposes a novel ECA-TCN (Efficient Channel Attention + Temporal Convolutional Network) paired with OCSVM . (Section 4.1)	1.0: Achieves an AUC of 96% and provides comparative analysis against four other deep learning algorithms. (Abstract, Conclusion)	5.0	Include
90	Tao et al. (2025) Test-Time Teaining	1.0: Utilizes the benchmarked CMU CERT v4.2 Insider Threat Dataset. (Section 5.1)	0.5: Defines a 28-day time window for data modeling, but lacks detailed technical explanation of specific	0.5: Identifies multi-source behavioral attributes (Logon, Email, File, etc.) but treats them as	1.0: Clearly explains the hybrid TTT-ECA-ResNet structure, including residual blocks and test-time training logic. (Section 4.1)	1.0: Comprehensive results provided (AUC 98.75%, F1 96.81%) with comparative analysis against CNN,	4.0	Include

			normalization or encoding types. (Section 5.1, 5.5.1)	sequence inputs rather than defining manual behavioral rules. (Section 5.2, Table 1)		LSTM, and GRU. (Section 5.3, 5.4)		
91	Tian Zhihong et al. (2024) Dempster-Shafer Theory	1.0: Uses the benchmarked CERT Insider Threat Dataset v6.2. (Abstract, Sec 4)	1.0: Defines vectorization, cleansing, sampling, and dimensionality reduction steps. (Sec 3.1.1)	0.5: Identifies broad categories (Basic, Content, Traffic-based) but lacks specific behavioral rule logic (e.g., "accessing sensitive folders"). (Sec 3.1.1)	1.0: Detailed Attention-LSTM architecture and multichannel D-S fusion engine are well-explained. (Sec 3.1, 3.3)	1.0: Supported by clear metrics (95.79% DR, 95.47% Accuracy) and comparative analysis against six baselines. (Table 3)	4.5	Include
92	Tian Tian et al. (2025) specific threat scenarios	1.0: Utilizes the benchmarked CERT dataset (v6.2). (Abstract/Conclusion)	1.0: Extensive steps including daily serialization, One-hot encoding, TF-IDF for text features, and Isolation Forest for action outliers. (Section: Systematic Approach Design)	1.0: Identifies scenario-specific behaviors: sensitive themes in text, high-variance file operations, and off-hours equipment usage. (Section: Abnormal Operational Behaviors Analysis)	1.0: Employs a Transformer Encoder with a Multi-head Attention mechanism to capture long-distance dependencies. (Section: Sequence Analysis)	1.0: Clear metrics (Accuracy, Precision, Recall, F1) across three scenarios with comparison to LSTM, GNN, and GCN. (Results/Conclusion)	5.0	Include
93	Villarreal-Vasquez et al. (2023)	1.0: Utilizes a massive real-world dataset of 38.9 million events from a commercial network of 30 computers. (Sec 1 & 5)	1.0: Explicitly defines a transformation function $f_T(\cdot)$ and feature selection set F for categorical mapping. (Sec 4.2)	0.5: Identifies 6 core telemetry features (Actor, Action, Target, etc.) but focuses on low-level system events rather than high-level human behavioral rules. (Sec 5)	1.0: Provides a detailed description of the LADOHD framework, including a 3-layer LSTM encoder and BPTT settings. (Sec 4.3)	1.0: Reports high-fidelity metrics (TPR 97.29%, FPR 0.38%) and comparative analysis against HMM and EDR systems. (Sec 1 & 6)	4.5	Include
94	Wall & Agrafiotis (2021)	1.0: Utilizes the recognized CMU CERT r4.1 synthetic dataset for validation. (Section 5)	1.0: Defines daily behavior aggregation, a workday split, and a Z-score based discretization function (Processor formula) into seven intervals. (Section 5.2, 5.2.3)	1.0: Identifies specific features like "logging onto unusual PC" and "attachments sent" derived from threat theory. (Section 5.2.2)	1.0: Detailed technical explanation of BN structure learning (chi-squared) and linear-time inference optimizations. (Section 3, 4, 6.3)	1.0: Findings supported by AUC and ROC curves across various training lengths and threat types. (Section 7)	5.0	Include
95	Wang Cheng & Zhu (2022)	1.0: Uses the CMU (CERT) insider threat dataset alongside four other real-world datasets (Banking, Kyoto, Darknet, Gowalla). (Section IV-A)	1.0: Defines a detailed "Weight Smooth" function to normalize edge importance and a time-sequence split (time-based training/testing) to avoid "time-crossing." (Section III-B & IV-A)	0.5: Identifies core behavioral properties (Subject, Operation Type, Object, Host) but focuses on "internal associations" rather than manual expert-defined heuristics. (Section II-A)	1.0: Technically sound Behavioral Identification Graph (BIG) paradigm using an Event-Property Composite (EPC) model. (Section III-C)	1.0: Uses Standardized Partial AUC (SPAUC) to focus on the meaningful "head" of the ROC curve, providing a robust industry-standard evaluation. (Section IV-A)	4.5	Include
96	Wang & El Saddik (2023) DTITD	1.0: Evaluated on both the dense CERT r4.2 and the	1.0: Defines a complex pipeline including Tomek Links for down-sampling	1.0: Explicitly lists 6 categories of expert features: Work Hours,	1.0: Proposes DistilledTrans (a simplified Transformer encoder) and compares it with BERT,	1.0: Comprehensive results (Accuracy, Precision, Recall, F1,	5.0	Include

		sporadic/imbalanced CERT r6.2 datasets. (Section III-B, IV-B)	and BERT/GPT-2 for contextual data augmentation. (Section III-C)	PC ownership, Removable Media, File Type, Website category, and Email recipient. (Section III-D-3)	RoBERTa, and hybrid CNN/LSTM models. (Section III-F, III-H)	AUC) with comparison to 18 state-of-the-art baselines. (Tables 3, 4, 5)		
97	Wang Jiarong, * al. (2023) Deep Clustering	1.0: Utilizes the benchmarked CMU-CERT r4.2 dataset covering 1000 employees. (Sec 4.1)	1.0: Clearly defines entity extraction (Time, Host, User, Action, Magnitude) and discretization of action counts. (Sec 4.1)	0.5: Identifies 5 core log entities and mentions the 3 CERT scenarios, but relies on automated feature learning rather than new manual rules. (Sec 4.1)	1.0: Proposes a technically sound Encoder-Decoder (GRU-based) joined with a Deep Clustering network using SGD for optimization. (Sec 3.2)	1.0: Provides comprehensive metrics including Recall (95.11%), AUC (>90%), and comparative analysis against four baselines. (Sec 4.4, Table 2)	4.5	Include
98	Wang Jiarong et al. (2022) Fusion Model	1.0: Uses the benchmarked CERT r4.2 dataset, specifically isolating HTTP records. (Sec 4.1)	1.0: Defines bipartite graph mapping, inverse frequency weighting for domain importance, and Min-max normalization. (Sec 3.2.1, 3.3)	0.5: Identifies specific URL themes (job sites, competitors) but lacks a broader psychological or heuristic rule-set. (Sec 4.1)	1.0: Technically sound fusion of Individual (IUAD) and Peer-Group (PGAD) models using the Dominant Set framework. (Sec 3.1, 3.2)	1.0: Employs AUC and MMP (Minimum Misclassification Point) with comparison against 4 identification models. (Sec 4.3-4.5)	4.5	Include
99	Wang Zhi et al. (2024) FeD ITD	1.0: Uses the benchmarked CERT r6.2 dataset, simulating a large organization with 4,000 employees. (Section IV-B)	1.0: Detailed process including daily grouping, chronological sorting, numeric tokenization, and NLP-based data augmentation. (Section IV-B)	0.5: Identifies standard log attributes (activity time, logon, device, file, email) but focuses on automated pattern discovery over manual rules. (Section IV-B)	1.0: Technically robust framework combining LLMs (RoBERTa/XLNet), FedAvg, PETuning (LoRA/Adapter), and Deep CORAL. (Section III)	1.0: Extensive evaluation using Macro Average F1, AUC, and comparative analysis against 10+ baselines. (Tables 2, 8, 9)	4.5	Include
100	Wei Yichen et al. (2021) Unsupervised Learning	1.0: Uses the benchmarked CMU-CERT r6.2 dataset and KDD CUP 99 . (Sec 4.1, 5.1)	1.0: Defines per-user/per-day behavior serialization, one-hot encoding, and padding/truncation to average length. (Sec 4.1)	0.5: Identifies 164 behaviors but treats them as tokens for automated BiLSTM learning rather than manual heuristic rules. (Table 1, Sec 4.1)	0.5: Cascaded Autoencoders (CAEs) for purification and joint AE-GMM optimization. (Sec 3)	1.0: Findings supported by AUC metrics and filtering performance curves across varying anomaly ratios. (Sec 5)	4.0	Include
101	Wei Zhiyuan et al. (2024) E-Watcher	1.0: Uses the benchmarked CERT r4.2 dataset, specifically focusing on scenarios 1 and 2. (Sec 4.1)	1.0: Defines time-interleaving merging, One-Hot Encoding, and variance threshold-based filtering. (Sec 3.1, 3.2)	0.5: Utilizes Shannon Entropy and a 3- σ empirical rule for thresholds, but focuses on statistical significance over specific manual behavioral rules. (Sec 3.3.1)	1.0: Technically sound "E-Watcher" hybrid framework combining Information Gain (statistical) and Local Outlier Factor (unsupervised ML). (Sec 3.3)	0.5: Reports 98.48% accuracy, but the comparative Table lacks baseline metrics for half of the listed studies. (Table 2)	4.0	Include
102	Wen et al. (2023)	1.0: Uses the real-world Enron email dataset, a benchmark for fraud and internal threat defense. (Sec 4.1.1)	1.0: Detailed filtering of non-Enron domains, removal of void content, and weekly aggregation to handle data sparsity. (Sec 3.1)	0.5: Focuses on sentiment polarity/depth as the primary indicator via the VADER lexicon. (Sec 3.2)	0.5: Uses Singular Value Decomposition (SVD) and Eigenvector Centrality; while sound, it is a lightweight unsupervised approach compared	0.5: Validates effectiveness via the Enron case study, mapping sentiment declines to the company's financial	3.5	Include

					to deep temporal models. (Sec 3.3)	crisis/bankruptcy. (Sec 4.2)		
103	Xiao Junchao et al. (2023) Graph Neural Network	1.0: Employs the benchmarked CERT r6.2 dataset (4,000 users). (Sec III-A)	1.0: Sophisticated use of Word2Vec for text, Min-Max normalization, and Visibility Graph (VG/HVG) construction. (Sec III-B, C)	0.5: Captures "contextual relationships" through graph edges but relies on learned weights rather than manual heuristic rules. (Sec III-C)	1.0: Technically advanced MEWRGNN combining R-GCN and CAN-GAT for multi-relational feature fusion. (Sec IV)	1.0: High accuracy (AUC 0.996) with detailed evaluation of detection delay and feature interpretability. (Sec V-C, F)	4.5	Include
104	Xiao Haitao et al. (2024) Unveiling shadows	1.0: Uses the benchmarked CERT dataset for effectiveness and robustness testing. (Sec 1)	1.0: Comprehensive chronological log merging, daily segmentation, and min-max normalization into 18x18 matrices. (Sec 3.2, 3.3.2)	0.5: Incorporates "Big Five" psychometric scores and specific PC types (e.g., supervisor's PC) as critical indicators. (Sec 3.2, 3.3.1)	1.0: High-complexity dual-stream architecture (Convolutional Attention + Transformer Encoder) for multi-dimensional learning. (Sec 3.1)	0.5: Demonstrates strong performance, but emphasizes that certain scenarios like "Intellectual Property Theft" remain difficult to detect. (Sec 6)	4.0	Include
105	Xiao Fengrui et al. (2025) SENTINEL	1.0: Validated on three major public datasets: LANL, CERT r6.2, and TWOS. (Sec V-A)	1.0: Innovative "Behavior Interaction Graph" (BIG) where hosts are nodes and logs are edges, reducing node count and computational complexity. (Sec IV-B)	0.5: Identifies specific topological patterns (In-Star, Out-Star, K-Path) as precursors to threats. (Sec III-B)	1.0: Advanced Spatio-Temporal Graph Neural Network (ST-GNN) with Edge Graph Attention (EGAT) and multi-timescale fusion. (Sec IV-C, D, E)	1.0: Superior performance (AUC 0.980) with detailed ablation studies on time granularity and feature importance. (Sec V-E, G)	4.5	Include
106	Ye Xiaoyun et al. (2025) Federated Learning	1.0: Utilizes two versions of the benchmarked CERT dataset (v4.2 and v5.2). (Sec: Dataset)	1.0: Highly technical preprocessing using the DeepInsight method (t-SNE + 64x64 pixel grid) to convert logs to images. (Sec: Methods)	0.5: Integrates "Big Five" personality scores and LDAP profiles, but focuses on CNN-based automated feature extraction. (Sec: Methods)	1.0: Combines a Federated Learning framework (privacy-preserving) with a CNN architecture for spatial pattern learning. (Sec: Contributions)	1.0: Reports significant improvements in accuracy and recall compared to the baseline FedAT model. (Sec: Results)	4.5	Include
107	Yi & Tian (2024)	1.0: Uses the benchmarked CERT r4.2 dataset (1,000 users, 500 days). (Sec 4.1)	1.0: Employs Z-score/Min-Max normalization and PCA for dimensionality reduction of the feature space. (Sec 3.1, 3.3.2)	0.5: Utilizes psychographic information and daily/weekly temporal granularity as behavioral markers. (Sec 4.1, 4.2)	1.0: Technically sound hybrid architecture (KLI-XG) leveraging unsupervised scoring as a nonlinear transformation for supervised classification. (Sec 3.1)	0.5: Reports a high AUC (0.998) and 86.12% accuracy, though focused on a restricted 20% "investigation budget." (Sec 5)	4.0	Include
108	Yumlembam et al. (2025)	1.0: Benchmarked on two distinct versions of the CERT dataset (r5.2 and r6.2) for cross-scenario validation. (Sec V-A)	0.5: Defines activity encoding (Type x 24 + Hour) and standard scaling, but relies on simple replication for data imbalance rather than synthetic generation. (Sec V-B)	1.0: Highly detailed manual feature set including supervisor PC access, BCC email counts, and specific flag-word tracking. (Sec IV-B)	1.0: Complex, technically sound integration of Dual-GCN (Explicit/Implicit), Multi-head Attention, and Bi-LSTM. (Sec IV-D, E)	1.0: Comprehensive comparison with LAN, DeepLog, and Log2vec using AUC, DR, and FPR metrics. (Sec V-C)	4.5	Include
	Yusof et al. (2022)	1.0: Uses ground-truth production traffic (PCAP files) from a major Malaysian	0.5: Defines a Feature Selection Model based on distribution density functions (Beta, Gamma,	0.0: Features are purely network-layer (window size, frame length, delta time) and	0.5: Bayesian Network implementation used for probabilistic modeling and	1.0: Presents Accuracy, False Positive Rate, and error measurements	3.0	Exclude

		healthcare provider and simulated attacks. (Sec 3.1)	MLE), but lacks standard ML scaling/encoding details. (Sec 4.1)	do not include human/expert behavioral rules for insiders. (Tab 3)	predictive analytics of Network Traffic. (Sec 3)	(AIC/BIC) compared against SVM and k-NN. (Sec 4.1)		
109	Zhang et al. (2021)	1.0: Uses both CERT v4.2 and v6.2 benchmark datasets. (Sec 4.1)	1.0: Detailed TF-IDF entity embedding and avg-TI normalization to (0,1) for disparate log attributes. (Sec 3.2)	0.5: Defines 10 behavioral dimensions (Table 1), though these are largely direct mappings of log attributes rather than complex derived expert rules. (Sec 3.2)	1.0: Innovative use of Over-Bootstrap sampling and a self-supervised pretext task (User ID) to regularize a Bagging-based LSTM ensemble. (Sec 3.3, 3.4)	1.0: Reports high AUC (99.2% for v4.2) and compares performance against state-of-the-art RNN/CNN models. (Sec 4.3)	4.5	Include
110	Zhu et al. (2024) AUTH	1.0: Uses the benchmarked CERT dataset, specifically noting its alignment with industrial ERP system logs. (Sec IV-A)	1.0: Clearly defines session-based segmentation (login to logout) and the separation of event frequency vs. time features. (Sec III-C)	0.5: Identifies key behavioral types (Email, Device, File) and the importance of working hours vs. late night, but relies on automated scoring. (Sec III-C)	1.0: Technically advanced TL-AAE combining TCN and LSTM with Generative Adversarial Theory for latent space regularization. (Sec III-D)	1.0: Reports AUC (0.932) and EER (0.146) with extensive comparison against OCSVM, IF, and DAGMM. (Sec IV-D)	4.5	Include

- **Total Identified for Full-Text:** 117
- **Total Excluded (Score < 3.5):** 7
- **Final Primary Studies (Include):** 110